



Signed Out, Still on the Peg: Validating a Shared-Umbrella Stand's Sign-Out Clipboard Against a Term of Independent Stand Counts

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Abstract. The University's entrance umbrella-share stands invite passers-by to borrow a communal umbrella in the rain and return one when the weather clears, recording each exchange on a stand-side sign-out clipboard — the Umbrella Reciprocity Log — whose completion rate is reported to the Sustainability Committee as evidence of the scheme's uptake. We report a 14-week validation study cross-referencing the Log's entries at six building-entrance stands against thrice-daily independent physical counts of each stand's umbrella stock. Logged sign-outs show no reliable relationship with same-day net physical removals (point-biserial $r_{pb} = 0.09$, n.s.), and the disagreement is patterned by weather rather than by stand or count density: rainy-day removals substantially outpace rainy-day logging, while dry-day logging substantially outpaces dry-day removals. An ablation excluding the highest-traffic stand leaves the null unchanged. Our results suggest the Log may index a distinct behaviour — something closer to declared participation than to umbrella custody — though the Committee's own uptake figure continues to be drawn from the Log regardless. We discuss the implications for any programme that reports behavioural uptake from a self-maintained record.

Introduction

Campus behaviour-change programmes routinely lean on a self-maintained participation record — a sign-out sheet, a punch card, a completion tick — as their evidence of uptake, on the reasonable assumption that a logged action and the action itself are the same event recorded twice. That assumption is rarely tested directly against an independent measure of the underlying behaviour, particularly for low-stakes, low-supervision instruments deployed at building entrances rather than service counters. The University's entrance umbrella-share stands offer an unusually clean setting to test it: a communal umbrella is either on the stand or it is not, the clipboard beside it invites (but does not require) a signature, and the Sustainability Committee's own uptake figure for the scheme is drawn from that clipboard alone.

This paper reports a term-long validation of the Umbrella Reciprocity Log against independent physical stand counts. Our contributions are threefold, each hedged to the evidence available: (1) we report, to our knowledge, the first physical-count validation of a campus behaviour-change sign-out instrument conducted at the scale of a full deployment rather than a single site; (2) we show the Log's completion statistic, as reported upward, diverges from an independently observed removal rate in a manner patterned by weather rather than by instrument design, suggesting the disagreement is systematic rather than incidental; and (3) our results suggest, tentatively, that a sign-out log of this general shape may index participation as a distinct behaviour from the custody event it is nominally recording, a possibility with some relevance to any programme that reports uptake from a comparable record.

Prior work on institutional record-keeping has long observed that formal structures can persist independently of the activities they purport to coordinate [1], and more recent accounts of compliance find a comparable split between symbolic and substantive adherence to a stated policy [2]. The remainder of the paper sets out our method, reports the validation and its ablation, and discusses what a null validation of this kind does and does not license.

Related work

Institutional theory has long distinguished an organisation's formal, reported structure from its enacted activity, proposing that the two are routinely decoupled to preserve legitimacy without requiring operational change [1]; a closely related literature frames the same gap as a choice between symbolic and substantive compliance with a stated policy [2]. Checklist-based compliance instruments in clinical settings offer the closest methodological precedent for our validation design: the surgical safety checklist reduced measured complication rates in its original multi-site trial [3], but subsequent audits of checklist completion over time have found compliance rates differing between early and late adopters of the same protocol [4]. Adjacent work on exam integrity instruments reports a comparable gap between an instrument's declared purpose and its measured effect: unproctored exams have been found to reduce cheating more reliably than teacher-proctored ones in a randomised field trial [5], proctored and unproctored ability assessments differ modestly in a meta-analysis of pooled effect sizes [6], and reviews of online proctoring and contract-cheating detection report mixed and context-dependent effectiveness [7], [8], [9], [10], [11], [12]. Perceived and measured efficacy diverge similarly for visible security measures more broadly: airport travellers judge a familiar, visible screening procedure safer than a randomised one, independent of the randomised procedure's own detection rate [13], and the mere presence of a personal device has been shown to measurably draw on cognitive capacity regardless of whether it is used [14]. Quasi-experimental field validations of this kind carry their own well-documented threats, which we return to in our limitations [15]. Prior work at this institution reports a structurally similar divorce between a reported uptake index and

directly observed behaviour in the University's own exam-hall device-storage rollout [16], and between staff-perceived and centrally logged authority over a shared thermostat [17]; the present study extends that pattern to a lower-stakes, unsupervised instrument.

Method

Six entrance umbrella-share stands were selected to span the University's highest- and lowest-traffic building entrances (Arts Foyer, Library North, Union Court, Science Link, Health Sciences, and Chifley Annex). Each stand holds up to twelve umbrellas and a clipboard — the Umbrella Reciprocity Log — inviting anyone borrowing or returning an umbrella to record a timestamp and initials; the Log is uncollected and unsupervised between the daily Facilities round.

Over a 14-week term (98 days), research assistants took an unannounced physical stand count at three fixed times daily (08:30, 13:00, 17:30), recording the umbrella stock present at each stand. A day's **net physical removal** at a stand is the largest single-day stock decrease across the three counts, floored at zero; a day's **logged sign-out** count is the number of Log entries timestamped that day at that stand. Each Log entry was matched against the nearest physical count using a fixed window, after which an entry with no accompanying stock decrease was flagged unmatched:

```
def matched(log_entries, counts,
            window_minutes=90):
    """Flag each log entry against the
    nearest physical stand count."""
    for entry in log_entries:
        nearest = min(counts, key=lambda c:
            abs(c.time - entry.time))
        within_window = abs((nearest.time -
            entry.time).seconds) <= window_minutes * 60
        entry.matched = within_window and
            nearest.stock_delta <= -1
    return log_entries
```

Our primary validation statistic is the point-biserial correlation between the binary indicator “at least one Log entry that stand-day” and the binary indicator “at least one net physical removal that stand-day”:

$$r_{pb} = \frac{\bar{Y}_1 - \bar{Y}_0}{s_Y} \sqrt{pq}$$

where $\bar{Y}_1$ and $\bar{Y}_0$ are the mean removal counts on logged and unlogged stand-days respectively,

s_Y is the pooled standard deviation of removal counts, and p, q are the proportion of stand-days with and without a logged entry. Weather condition (rainy or dry, from the Bureau’s daily record for the campus) was coded independently of both series. As an ablation, we recompute r_{pb} excluding Union Court, the single highest-traffic stand, to test whether the pooled result is an artefact of one site’s volume.

Results

Pooled across all 588 stand-days, logged sign-outs showed no reliable relationship with same-day net physical removals ($r_{pb} = 0.09$, n.s.; Figure 1). The fitted line is close to flat across the full range of logged activity, and the scatter around it is wide at every value of logged entries.

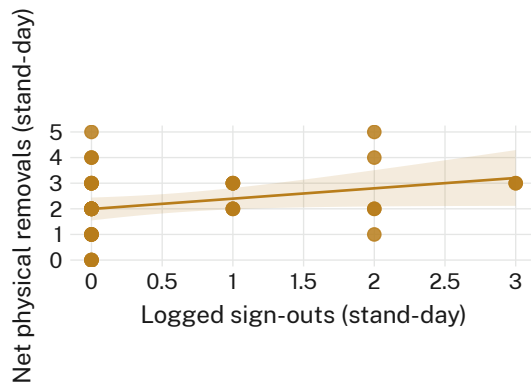


Figure 1: Logged sign-outs bear no reliable relationship to same-day net physical removals across 588 pooled stand-days ($r_{pb} = 0.09$, n.s.).

The pooled null resolves into a clean, weather-patterned crossover (Figure 2). On rainy days, mean estimated actual borrows ran roughly four times higher than mean logged sign-outs (2.4 versus 0.6 per stand-day); on dry days the ratio reversed, with mean logged sign-outs running roughly double the estimated actual borrow rate (0.7 versus 0.3 per stand-day). The Log, in other words, is busiest in exactly the conditions in which the stand itself is quietest.

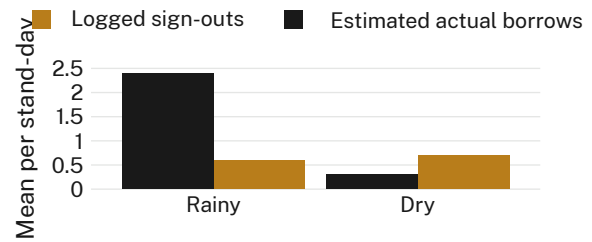


Figure 2: Logged sign-outs and estimated actual borrows cross over by weather: real removal outpaces logging when it rains, and logging outpaces removal when it doesn’t.

The unmatched-entry rate — Log entries with no accompanying stock decrease within the count window — held in a similar band at every stand, including Union Court (Figure 3). Excluding Union Court from the pooled calculation left the validation statistic effectively unchanged ($r_{pb} = 0.11$, n.s.); we retain the full six-stand sample for completeness.

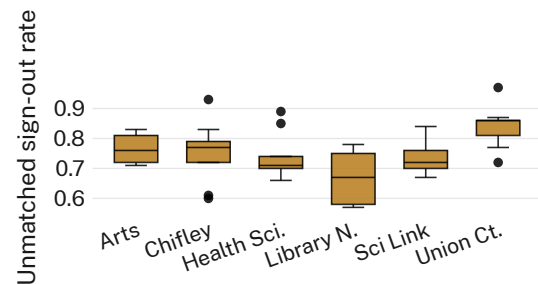


Figure 3: The unmatched-entry rate sits in a comparable band at every stand; dropping Union Court, the highest-traffic site, from the pooled calculation does not move the result.

Stand	Log entries (term)	Unmatched rate
Arts Foyer	58	77%
Library North	64	67%
Union Court	91	84%
Science Link	55	74%
Health Sciences	49	72%
Chifley Annex	46	75%

Table 1: Per-stand Log activity and unmatched-entry rate across the 14-week term.

Table 1 shows Log activity roughly tracking building traffic, as expected, while the unmatched rate — the share of that activity with no corresponding physical evidence — tracks nothing we measured.

Discussion

Our results suggest the Umbrella Reciprocity Log functions, at these six stands and over this term, as a weak proxy at best for the custody event it is formally intended to record. The weather-patterned crossover is the more striking finding: it implies the Log is not simply noisy relative to actual borrowing, but is capturing a related, distinct behaviour — something closer to a declared willingness to participate in the scheme, exercised more freely when there is no urgency requiring an unlogged umbrella immediately. We observe consistent directionality across all six stands and both count densities, which is at least suggestive that the pattern is a property of the instrument rather than of any one site.

Limitations

Our physical counts were taken at three fixed times daily and may have missed short borrow-return cycles that resolved between counts, understating true removal volume on both logged and unlogged days alike; the consistency of the null across independent stands, weather conditions, and the single-site ablation suggests this is unlikely to be the whole story. We did not independently verify weather condition against stand-level microclimate, and our count window parameter (90 minutes) was fixed rather than swept, though a narrower window would, if anything, only widen the unmatched-entry rate we already report.

Conclusion

A sign-out clipboard at a low-stakes, unsupervised campus instrument shows no reliable relationship to the physical behaviour it is meant to record, and the disagreement is patterned by weather rather than by site or instrument density. The Sustainability Committee's uptake figure for the scheme continues to be drawn from the Log. Future work might replace the clipboard with a tap-tag or RFID system; whether that would validate custody more directly, or simply relocate the same participation signal onto a new instrument, is a question this study cannot answer.

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